

Yallah Nsafro

Feasibility Report and Software Requirements Specification

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Software Engineering — Version 0.1

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Outline

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- 3 Software Requirements Specification

Yallah Nsafro — Project Description

What is it?

A **web app** that turns a trip request into an **editable day-by-day trip plan** with a **budget estimate**.

The traveler provides

- Destination and departure city
- Dates or trip duration
- Number of travelers
- Budget
- Interests

The app returns

- Day-by-day itinerary (editable)
- Suggested flights, hotels, restaurants, activities
- Total budget with category breakdown
- Links to providers to check or book

Project Description — MVP Focus

Data sources

Travel options come from approved providers/APIs where available. If live data is unavailable, the app uses clearly labeled sample data or user-entered estimates.

First version (MVP) = planning and recommendations

- Plan, suggest, estimate, save, link to providers
- **No** payment processing
- **No** ticket issuance
- **No** booking inside the app

Key principle

Every price is labeled **Live**, **Estimate**, **Sample** or **User-entered**, with its retrieval time.

Feasibility — Goal and Scope

Goal

Generate an editable day-by-day plan with suggested options and a **labeled** budget estimate.

Recommended MVP

Planning + recommendations + outbound provider links. Sample / user-entered data as fallback.

Legend

V = verified in official provider docs (5 Oct 2026)
validate

A = assumption to validate

Findings

- **V** Amadeus Self-Service APIs were decommissioned on 17 Jul 2026 (Enterprise contract only): flights need another source.
- **V** Booking.com Demand API is open to affiliate partners via an agreement and an account manager.
- **V** Google Places API (New) covers restaurants and attractions.
- **A** The team can build a web app (e.g. React + Node) and integrate REST APIs.

Verdict

Feasible with conditions (provider access).

Feasibility — Market and Economic

Market / user

A Target users: students and young travelers planning budget trips.
Benchmark of competing planners and user survey: **to complete**.

Economic

- **V** Google Maps Platform: free monthly calls per SKU (10K Essentials, 5K Pro, 1K Enterprise), then pay-per-use.
- **V** Pricing model changed in Mar 2025.
- **A** Hosting, AI (LLM) tokens, flight/hotel data fees: not quoted yet. Cache results and cap calls per user.

Verdict

Low fixed cost, variable usage cost.

Feasibility — Operational, Schedule, Legal

Operational

A Fits a small student team; needs quota monitoring and stale-price handling.

Schedule (assumption)

W1–2 elicitation + SRS · W3–4 design + prototype · W5–8 core features · W9–10 provider integration + fallback · W11 testing · W12 release

Legal / contractual

V Provider agreements govern display, caching and attribution. No legal conclusion drawn: confirm provider terms and privacy law before launch.

Feasibility — Main Risks

Risk	Mitigation
Provider access denied or delayed	Ship labeled sample / user-entered mode
Prices and availability change	Label each amount + show retrieval time
Coverage gaps by city	“Limited data” notice; manual additions
Misleading budget estimates	Show ranges, never a guarantee

Proceed with conditions

- 1 Secure at least one flight and one hotel source, or launch in estimate-only mode.
- 2 No unlabeled prices: provenance + timestamp everywhere.
- 3 Re-check provider terms and pricing before build and before release.

Sources

phocuswire.com/amadeus-shut-down-self-service-apis-portal-developers
developers.booking.com/demand/docs/accommodations/accommodation-quick-guide
mapsplatform.google.com/pricing

Purpose

Specify the requirements of Yallah Nsafo, a web app that builds editable day-by-day trip plans with budget estimates.

Scope

Will: collect trip inputs, generate and edit an itinerary, suggest flights, hotels, restaurants and activities, compute a budget, save plans, link to providers.

Will not: process payments, issue tickets, or book inside the app.

Definitions

Provenance label: Live, Estimate, Sample or User-entered. **Provider:** external travel data API. **MoSCoW:** M Must, S Should, C Could, W Want later.

SRS — Overall Description

Product

Standalone web app consuming third-party provider APIs. Main functions: trip creation, preferences, itinerary, suggestions, budget, saving, provider links.

Users

Traveler (primary, no technical skill). **Admin** (optional, monitors data sources).

Constraints and assumptions

- Provider terms limit caching/display; provider access is not guaranteed.
- No payments in the MVP (Minimum Viable Product).
- At least one approved provider per category, or fallback data.
- Prices can change at any time.

SRS — Functional Requirements (1/2)

ID	The system shall...	Pri	Status	Target
FR1	let the traveler create a trip with destination, departure city, dates or duration, number of travelers and budget	M	Proposed	V1
FR2	capture interest preferences	M	Proposed	V1
FR3	generate a day-by-day itinerary	M	Proposed	V1
FR4	let the traveler add, remove, reorder and edit itinerary items	M	Proposed	V1
FR5	suggest flights, hotels, restaurants and attractions/activities	M	Proposed	V1
FR6	compute a total budget with per-category breakdown	M	Proposed	V1

SRS — Functional Requirements (2/2)

ID	The system shall...	Pri	Status	Target
FR7	save and reopen a trip plan	M	Proposed	V1
FR8	provide an outbound link to the provider for each option	M	Proposed	V1
FR9	label every price Live, Estimate, Sample or User-entered and show retrieval time for Live prices	M	Proposed	V1
FR10	fall back to labeled sample or user-entered data when a provider is unavailable	M	Proposed	V1
FR11	reject invalid inputs (e.g. end date before start date) with a clear message	S	Proposed	V1

SRS — Non-Functional Requirements

ID	The system shall...	Pri
NFR1	generate a 7-day itinerary within 30 s in 95% of test runs	M
NFR2	display correctly from 360 px screen width without horizontal scrolling	M
NFR3	meet WCAG AA contrast and keyboard navigation on main screens	S
NFR4	use HTTPS, keep API keys server-side and store only data needed for saved plans	M
NFR5	let a user delete their saved plans	S
NFR6	let a first-time user create a plan in under 10 minutes without training	S
NFR7	remain usable (fallback mode) during a provider outage	M

SRS — Acceptance Criteria and MVP Boundary

Acceptance criteria

- 1 100% of displayed prices carry a provenance label (FR9).
- 2 Category totals equal the grand total (FR6).
- 3 Edits persist after page reload (FR4, FR7).
- 4 With the provider disabled, a full plan is still produced from labeled data (FR10).

Outside the MVP

Payments, ticket issuance, in-app booking, price alerts, collaborative plans.

Thank you

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